

Siyuan (Sylvester) Zhang

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EDUCATION

Columbia University

MS in Mechanical Engineering (Robotics and Control)

August 2024 ~ Jun 2026(Expected)

Department of Mechanical Engineering

- **Advanced Master Research Program**, Advisor: Prof. Hod Lipson

Sun Yat-sen University

BE in Aeronautical and Astronautical Engineering

August 2020 ~ Jun 2024

School of Aeronautics and Astronautics

RESEARCH INTEREST

Soft and Bio-inspired Robotics, Modular and Distributed Robotic Systems, Wearable and Assistive Exoskeleton, Robotic Material

PUBLICATION

- **Zhang, S., Lipson, H.** (2025). Robotic Mitosis: Engineering Self-Reproduction from 2D Lattices into 3D Morphogenetic Structures. *Manuscript in Preparation*.
- Zhang, Y.*, **Zhang, S.*, Lipson, H.** (2025). Design and Development of a Human-like Lip-Sync Robot Mechanism and Control Strategy. *Manuscript in Preparation*. (*Co-first authors)
- **Zhang, S.**, Zhang, Y., Lyu, J., & **Agrawal, S. K.** (2025). From Structural Design to Dynamics Modeling: Control-Oriented Development of a 3-RRR Parallel Ankle Rehabilitation Robot. *Manuscript in Preparation*.
- Zhao, Y., Zhu, J., Zhang, J., **Zhang, S.**, Shao, M., Chai, Z., ... & **Zhang, J.** (2025). Enhancing Grasping Diversity with a Pinch-Suction and Soft-Rigid Hybrid Multimodal Gripper. *IEEE Transactions on Robotics*.
- Ma, K., Zhang, J., Sun, R., Chang, B., **Zhang, S.**, Wang, X., ... & **Zhang, J.** (2024). Synergizing Structural Stiffness Regulation with Compliance Contact Stiffness: Bioinspired Soft Stimuli-Responsive Materials Design for Soft Machines. *Advanced Engineering Materials*, 26(10), 2400461.
- Zhao, Y., Zhang, J., **Zhang, S.**, Zhang, P., Dong, G., **Wu, J.**, & **Zhang, J.** (2023). Transporting dispersed cylindrical granules: An intelligent strategy inspired by an elephant trunk. *Advanced Intelligent Systems*, 5(10), 2300182.

RESEARCH EXPERIENCE

Design of An Intelligent Self-Reproducing Robot

Jan 2025 ~ Present

Advisor: Hod Lipson

Columbia University

- Developed a magnetically coupled, servo-actuated triangular module system capable of autonomous self-reproduction under positional uncertainty, achieving stable 3D-2D shape morphing and tetrahedral assembly.
- Engineered a new reproduction strategy reducing the minimal replicating unit from 4-module to 2-module “cells,” enabling multi-generation replication and robust attachment/detachment through a distributed communication protocol with minimal sensing.

Development of a 3-RRR Parallel Ankle Rehabilitation Robot

Jan 2025 ~ Present

Advisor: Sunil K. Agrawal

Columbia University

- Developing a wearable 3-RRR parallel mechanism to improve portability and motion freedom in ankle rehabilitation robot compared with traditional design.
- Designed and simulated inverse kinematics and Jacobian-based control, enabling precise motion tracking for multi-axis ankle support. Integrating therapist-inspired assistance/resistance modes, currently under validation for rehabilitation.

Pinch-Suction and Soft-Rigid Hybrid Multimodal Gripper

Apr 2023 ~ Apr 2024

Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang

Sun Yat-sen University

- Designed a soft-rigid hybrid gripper with multiple manipulation modes, enabling effective handling of objects of varying sizes and weights to address the limited adaptability of conventional grippers.
- Integrated gripping and suction mechanisms to increase stability and controllability during object manipulation.
- Enabled grasping diversity across weight (0.2 g–10 kg), fragility (jelly to aluminum), scale (0.46 mm–0.55 m), and shape (poorly pinchable/sackable), outperforming conventional grippers.

Bioinspired Stiffness-Regulating Materials for Soft Machines

Jun 2023 ~ Jan 2024

Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang

Sun Yat-sen University

- Developed elastomer-SMA composite structures with Joule-heat stiffness regulation to address the compliance-load trade-off in soft robotics.
- Characterized dynamic stress-strain responses and rapid stiffness switching through systematic material testing.

- Applied the material to adaptive grippers and wearable prototypes, enabling enhanced load bearing and versatility.
- Elephant Trunk Inspired Soft Gripper

May 2022 ~ Jun 2023

Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang

Sun Yat-sen University
- Proposed and prototyped a pneumatic gripper inspired by elephant trunk mechanics to tackle the challenge of manipulating irregular and granular object.
 - Optimized actuation pressures and deformation profiles through ANSYS and SOLIDWORKS simulations.
 - Achieved >90% grasp success rate and 50% faster operation time compared to baseline grippers in experimental validation.

PROJECT EXPERIENCE

Soft Robotic & Neural Interface Material

Aug 2025 ~ Present

Advisor: Prof. Chaoqun Dong

Columbia University

- Performing finite element simulations of soft neural interface devices, investigating strain localization, viscoelastic deformation, and interfacial adhesion to guide structural optimization of stretchable robotic materials.
- Designed and fabricated customized experimental platforms and mounting fixtures to improve the reproducibility and precision of soft-material testing.

Design and Gait Optimization of a Quadruped Spider Robot for Robotics Studio

Sep 2024 ~ Dec 2024

Advisor: Prof. Hod Lipson

Columbia University

- Designed a 4-legged robotic system with 2 DOF per leg, powered by 8 servomotors (240° range) and controlled via a Raspberry Pi. Created a parametric CAD model in SOLIDWORKS, converted it to a URDF file, and optimized the robot's gait using PyBullet simulations and parallel hill climber algorithms.
- Implemented control algorithms in Python to actuate servos, enabling sim-to-real to optimize walking patterns.
- Built and tested the prototype operating 3D printing for rapid iterations, attaining a final walking speed of 29 cm/s.

Design of a Bionic Flapping-Wing Robot Inspired by Birds

Sep 2022 ~ Dec 2023

Advisor: Prof. Zhenbo Lu

Sun Yat-sen University

- Developed a lightweight, foldable, and low-noise flapping-wing robot inspired by small birds, integrating principles of bionics.
- Devised and simulated the mechanical model using SOLIDWORKS and ANSYS, modifying structural performance and aerodynamics.
- Implemented a closed-loop control system to achieve stable and controlled flight.

Design of a Foldable, Adhesive Crawling CubeSat for Space Station Operations

Mar 2022 ~ Aug 2023

Advisor: Prof. Jinxiu Zhang, Jianing Wu

Sun Yat-sen University

- Led a team of 5 in designing a spider-inspired CubeSat with foldable claws, mechanical legs, and detection modules for surface inspection, debris removal, and damage repair.
- Optimized mechanical structure through SOLIDWORKS and explored electrostatic adsorption for secure attachment in space environments.
- Attained 3rd prize in the IAF-CSA Space Universities CubeSat Challenge 2.0 for innovative design and engineering solutions.

TEACHING EXPERIENCE

Graduate Teaching Assistant | Columbia University: Robotics Studio | Advisor: Hod Lipson

2025 Spring

- Contributed to robotics teaching and curriculum development, integrating python algorithm/ embedded systems to support 30+ students in hands-on robotic projects across undergraduate and graduate levels.

PROFESSIONAL EXPERIENCE

Insta360

Jan 2024 ~ Mar 2024

Structural Design Engineer Intern

- Conducted requirements analysis to translate product specifications into feasible structural designs, ensuring functionality, manufacturability, and cost-effectiveness.
- Constructed and finalized detailed 3D models and mechanical drawings operating Creo.
- Collaborated on prototyping and testing of product structures, improving design iterations and efficiency.

TECHNICAL SKILLS

- Programming: Python, C, MATLAB, C++, ROS, Linux, Arduino IDE
- Software: Auto CAD, MATLAB, ANSYS, SOLIDWORKS, PTC Creo, CATIA, NX CAD
- Engineering Techniques: FEA, DFM, DFA, FA, DOE, GD&T
- Manufacturing Techniques: FDM 3D Printing, Stereolithography (SLA), Laser Cutting, CNC Machining